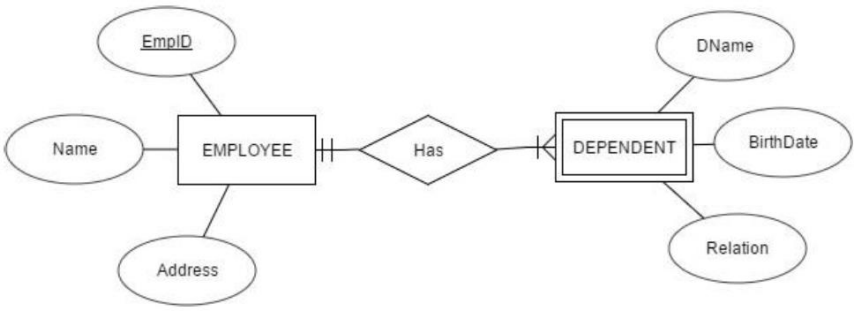
A. Example of ERD.

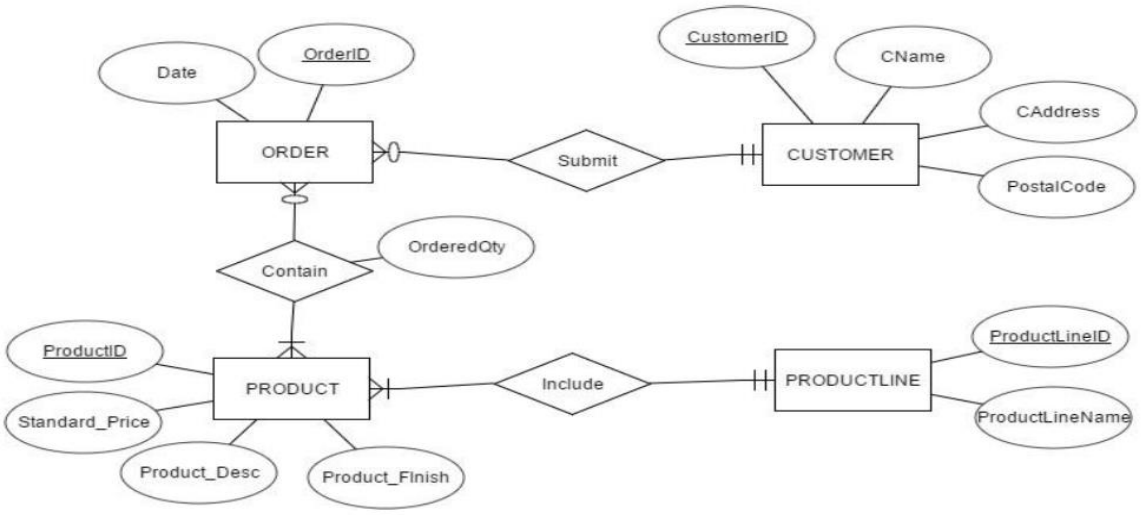
1. Entity type DEPENDENT related to EMPLOYEE. This entity type is used to keep track of the dependents of each employee via 1-N relationship. DEPENDENT entity has Name, Birthdate and Relationship. Two dependents of two distinct employees might have the same {Name, Birthdate, Relationship} values but they are totally different. They are identified as distinct only after consider the association with particular employee entities to which the dependents relates. So the key of the DEPENDENT might be {EmpID, DName} in which EmpID is the attribute from the strong entity type EMPLOYEE.



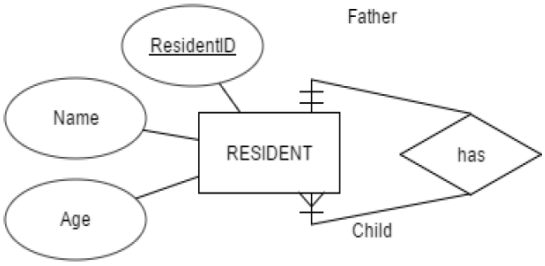
2. From a study of the business processes at Pine Valley Furniture Company, we have identified the following entity types. An identifier is also suggested for each entity, together with selected important attributes.

The company sells a number of different furniture products. These products are grouped into several product lines. The identifier for a PRODUCT is ProductID, whereas the identifier for a PRODUCTLINE is ProductLine_ID. We identify the following additional attributes for PRODUCT: Product_Description, ProductFinish, and Standard_Price. Another attribute for PRODUCTLINE is ProductLine_Name. A product line may group any number of products but must group at least one product. Each product must belong to exactly one product line.

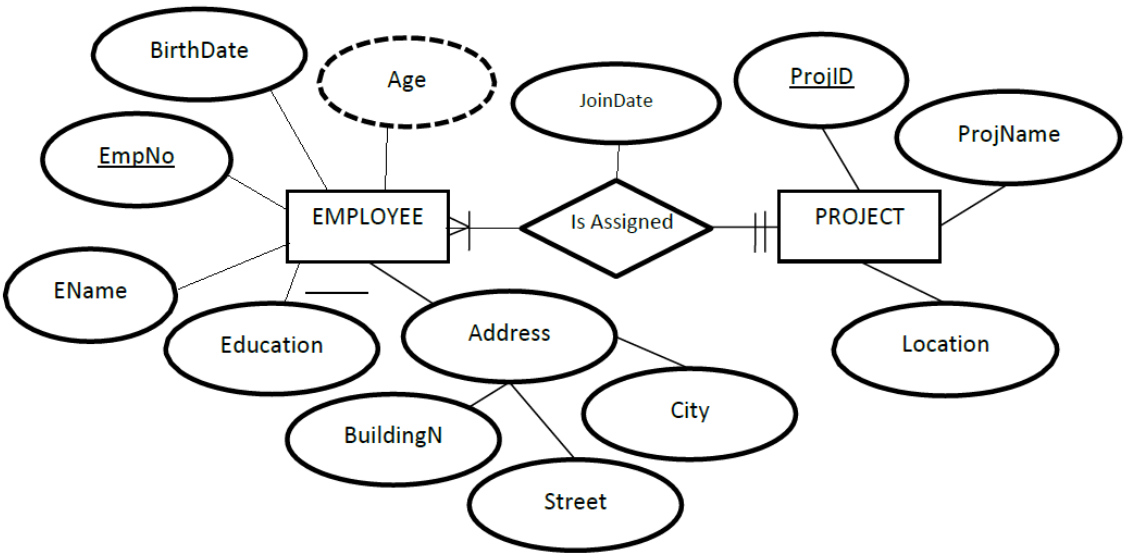
- ✓ Customers submit orders for products. The identifier for an ORDER is Order_ID, and another attribute is Order_Date. A customer may submit any number of orders, but need not submit any orders. Each order is submitted by exactly one customer. The identifier for a CUSTOMER is Customer_ID. Other attributes include Customer_Name, Customer_Address, and Postal_Code.
- ✓ A given customer order must request at least one product and product sold may not appear on any order or may appear on one or more order. The pine valley furniture wants to record order quantity for products associated with each order.



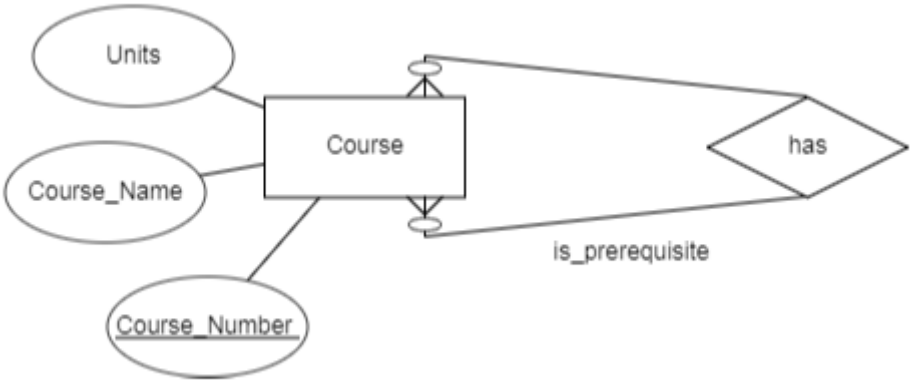
3. ROP interested to maintain information about citizens of Oman in its database. Attributes of RESIDENT include Resident_ID (identifier), Name, and age. ROP is interested to relate a father his/her children. Each father may have a child or children. Similarly, a child may have exactly one father.



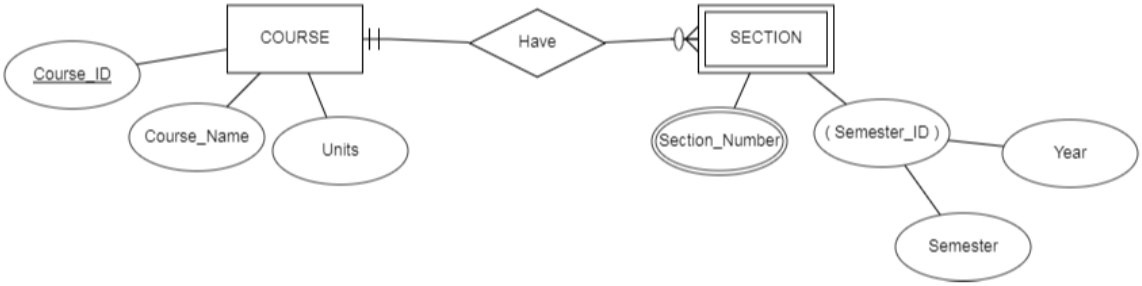
4. A Company has many employees and projects. The attributes of EMPLOYEE include EmpNo (identifier), Ename, BirthDate, Age (Which will get based on the BirthDate), Address (Address contain BuildingNo, Street, City) and Qualification (Some employees are having more than one Qualification). The Attributes of PROJECT include ProjID (identifier), ProjName and Location. Each employee can work only in one project. Each project is assigned to one or more employees. When an employee is assigned to a project the company records the JoinDate.



5. A university has a large number of courses in its catalog. Attributes of COURSE include Course_Number (identifier), Course_Name, and Units. Each course may have one or more different courses as prerequisites or may have no prerequisites. Similarly, a particular course may be a prerequisite for any number of courses or may not be prerequisite for any other course. Provide a good definition of COURSE. Why is your definition a good one?

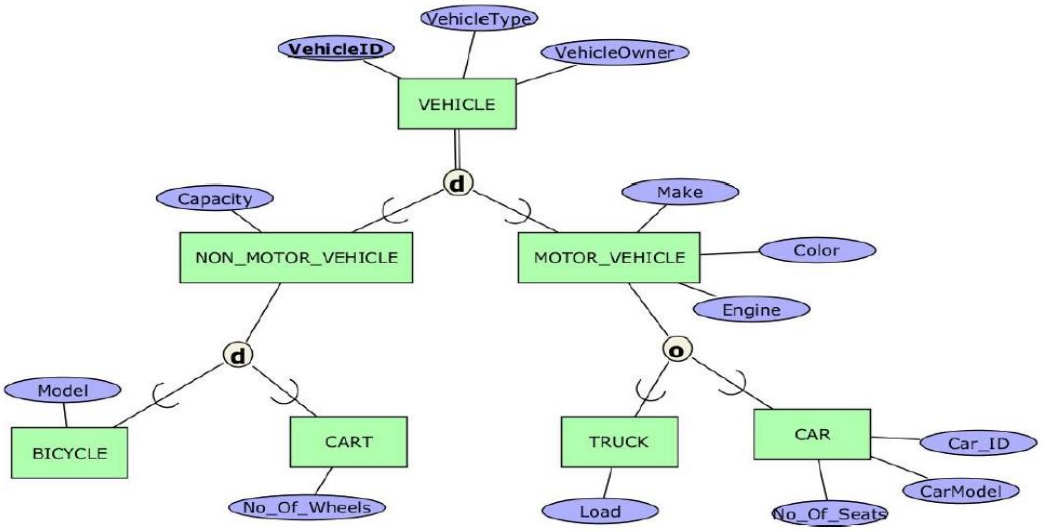


6. A college course may have one or more scheduled sections, or may not have a scheduled section. Attributes of COURSE include Course_ID, Course_Name, and Units. Attributes of SECTION include Section_Number and Semester_ID. Semester_ID is composed of two parts: Semester and Year. Section_Number is an integer (such as "1" or "2") that distinguishes one section from another for the same course but does not uniquely identify a section. How did you model SECTION? Why did you choose this way versus alternative ways to model SECTION?

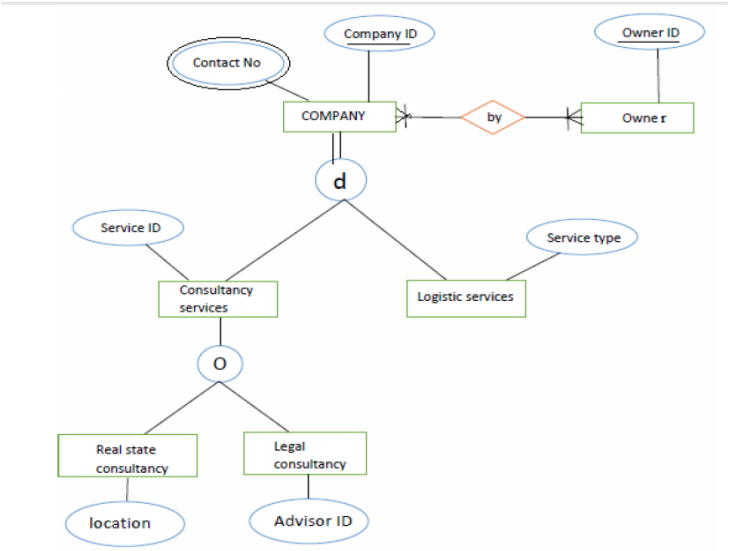


B. Example EERD

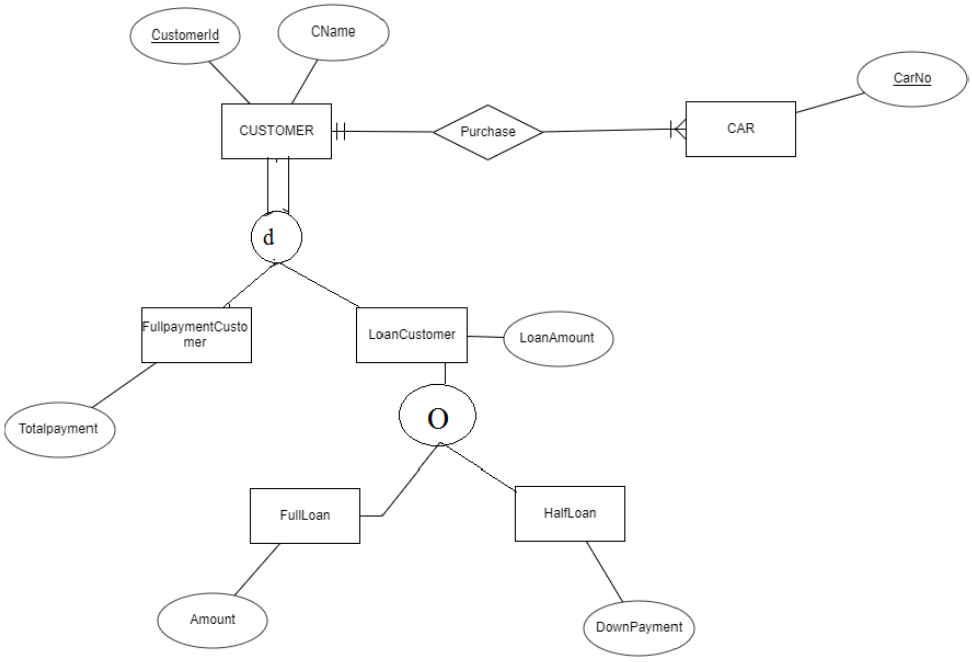
1. In Oman, different types of vehicles are used to transport the people and goods. Each Vehicle is identified by VehicleID and other details are VehicleType and VehicleOwner. All the types of vehicles are broadly classified into two; they are Non-Motor vehicles and Motor Vehicles. Motor Vehicles are not used as non-motor vehicles. The local attribute of Non-Motor vehicles is capacity while Make, Color and Engine are the local attributes of Motor Vehicles. There are different types of Non-Motor vehicles, of which some of them are Bicycle and Cart. A Bicycle cannot be used as a Cart. The local attribute of Bicycle is Model and that of cart is No_of_wheels. There are different types of Motor vehicles, of which some of them are Trucks and Cars. Sometimes Cars are used to transport as Trucks. The attribute of Trucks is Load while the car details are CarId, CarModel and No_of_seats.



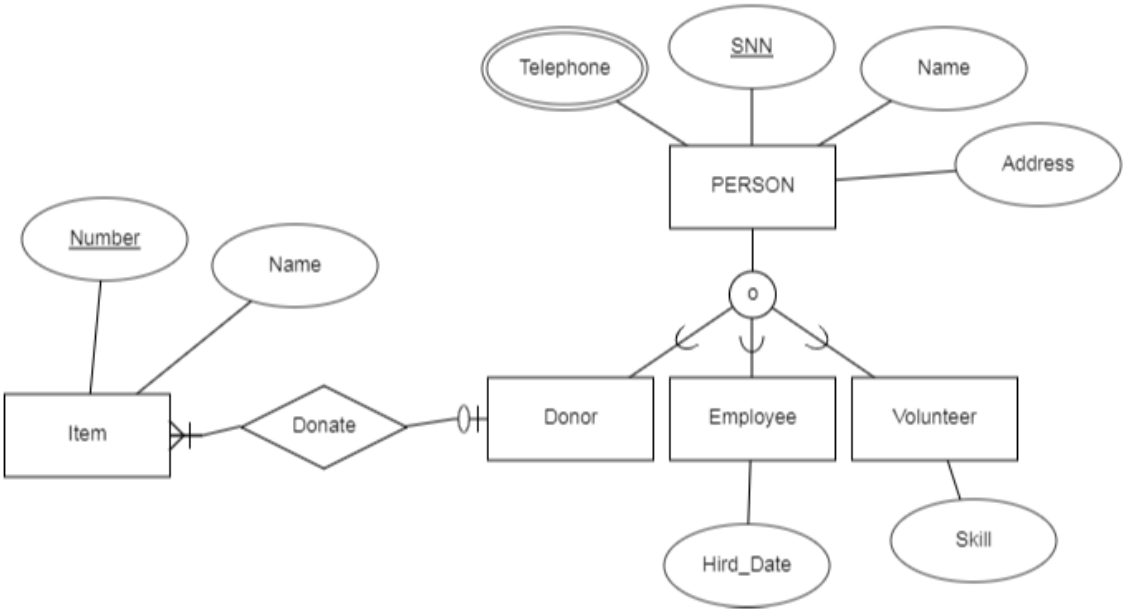
2. A company has a unique Company Id and Contact No (a company can have one or more contact number). The company is broadly classified in to Consultancy services and Logistic services. Consultancy services recorded its Service Id. Similarly Logistic services is recorded its Service type. Consultancy services cannot be serviced as Logistic. There are different type of Consultancy services some of them are Real State Consultancy and Legal Consultancy. Real State Consultancy recorded its Location. Legal Consultancy recorded its Advisor Id. Sometimes Real Consultancy can be serviced as Legal Consultancy. A company owned by one or more Owner also an owner can owned one or more companies. An Owner has an identifier Owner Id.



3. We want to create a database for a Nissan company that have customer who purchase Car. For this, we must store data about Customer and Car. Car has carno as an identifier. Company store customer details customerid (identifier) and Cname. Customer can purchase one more car but car will be purchased by exactly customer. All the types of Customers are broadly classified into two, they are FullpaymentCustomer and LoanCustomer. FullpaymentCustomer cannot purchase a car as LoanCust. The LoanAmount is recorded in the LoanCustomer while Totalpayment is recorded in the FullpaymentCustomer. There are different types in LoanCustomer some of them are FullLoan and HalfLoan. FullLoan can be recorded with amount. Sometimes HalfLoan are using as FullLoan. HalfLoan are recorded with DownPayment.

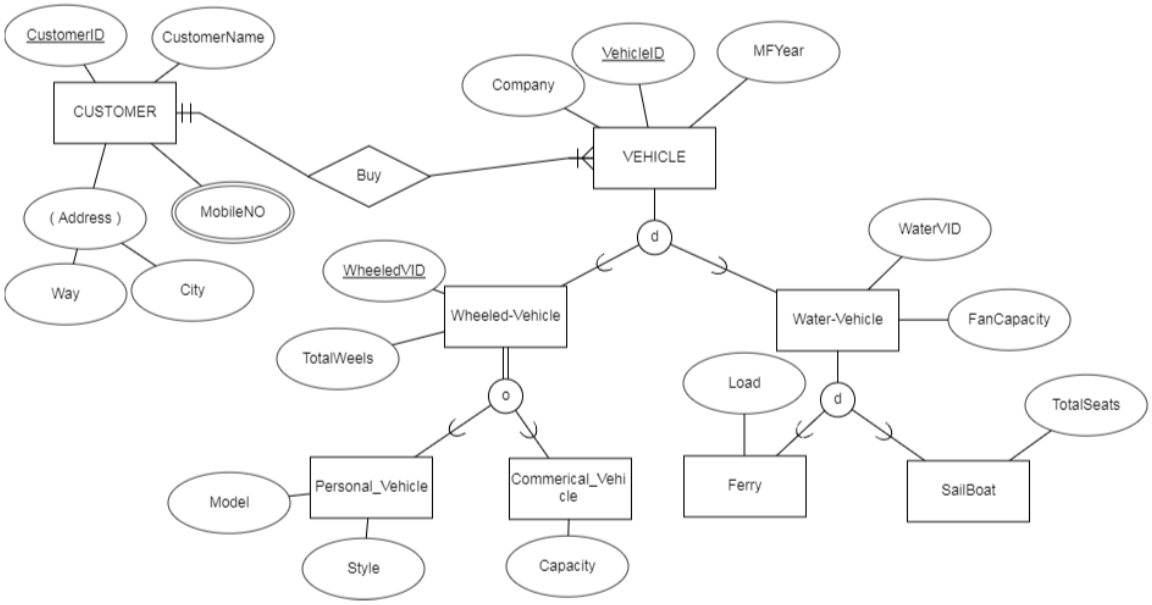


4. An Organization depends on a number of different types of persons for its successful operation. The organization is interested in the following attributes for all of these persons: SSN, Name, Address and telephone. A person may have multiple telephone numbers. Three types of persons are of greatest interest: employees, volunteers and donors. Employees only have a date hired attribute, and volunteers only have a skill attribute. Donors only have a relationship (named Donates) with an item that has number(identifier) and name. A donor must have donated one or more items, and an item may have no donors, or one donor. There are persons other than employees, volunteers, and donors who are of interest to the organization, so that a person need not belong to any of these three groups. On the other hand, at a given time a person may belong to two or more of these groups (for Example, employees and donors). Draw an EER diagram based on the requirements above. Write down assumptions (if any) you make in your answer.

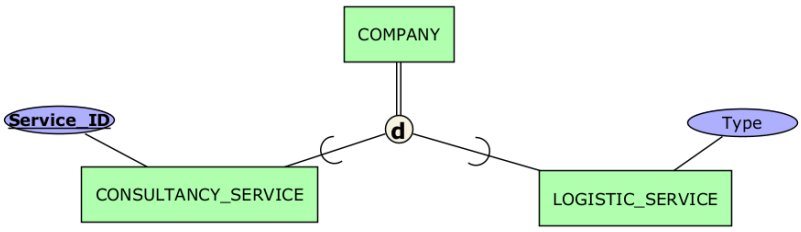


5. In Oman, Saud Bahwan Group is selling different brands and types of vehicles to the customers. A customer can buy either one or more than one vehicles, but each vehicle is owning by one and only one customer and a customer can refer to either none or more than one other customers to buy. A Customer can be identified with CustomerID and other details such as CustomerName, Address (which contain Way and City) and MobileNo (Some of the customers are having more than one Mobile Numbers). A vehicle can be identified with VehicleID, Company and MFYear. Based on the distinct features, the vehicles are classified into different types. Some of the types are Wheeled_Vehicles and Water_Vehicles. Water vehicles are not used as wheeled vehicles and vice versa. Wheeled_Vehicles are identified with WheeledVID and TotalWeels. Wheeled_Vehicles are classified into two only, they are Personnel_Vehicles and Commercial_Vehicles. Based on the requirement, Commercial_Vehicles can be used as Personnel_Vehicles. Personnel_Vehicles can be

recorded with Style and Model. Commercial_Vehicles can be recorded with Capacity. Water_Vehicles are idendified with WaterVID and FanCapacity. There are different types of Water_Vehicles; some of them are Ferry and SailBoat. Ferry cannot be used as a SailBoat and vice versa. Ferry can be recorded with Load. SailBoat can be recorded with TotalSeats.

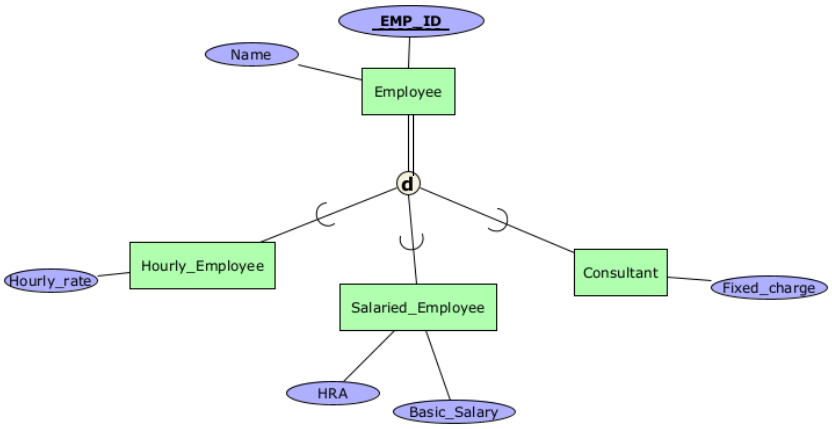


6. Draw an EERD for the following
   All Companies are classified in to Consultancy_Service and Logistic Service. A consultancy_service cannot function as Logistic_service and vice versa. Service_ID is recorded for consultancy_service .Type is recorded for Logistic_service

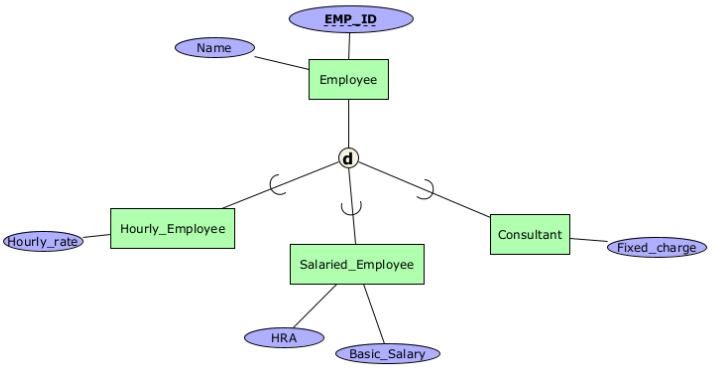


7. Draw an EERD for the following

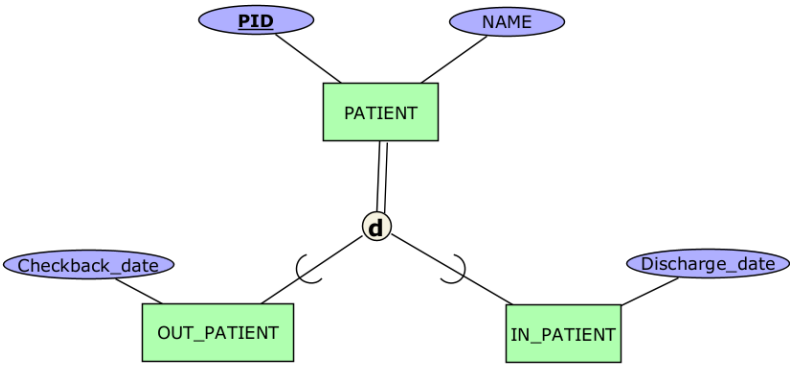
a)A company has different types of employees.ID and name will be recorded for all employees. A salaried employee will receive basic salary with house rent allowance (HRA). The employees working in hourly basis receive salary on hourly rate. The employees working as consultant will have fixed charges for their salary. An employee cannot belong to two types at a time.



b) A company has different types of employees. A salaried employee will receive basic salary with house rent allowance (HRA). The employees working in hourly basis receive salary on hourly rate. The employees working as consultant will have fixed charges for their salary. An employee cannot belong to two types at a time. Sometimes, there will be employees who are not belong to salaried employee, hourly_employee, and consultant.



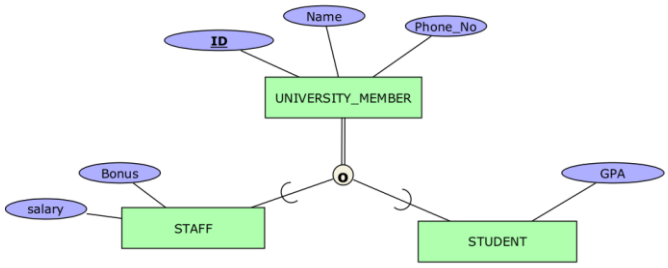
8. Write the business rules from the given model.



Hospital has two types of patients. Discharge date is recorded for every In_patient in the hospital. The Checkback date is recorded for every out_patient in the hospital. If a patient is a In_patient, then he/she cannot be an out_patient (a patient cannot be IN and OUT patient at a time).

9. Draw an EERD for the university system given below.

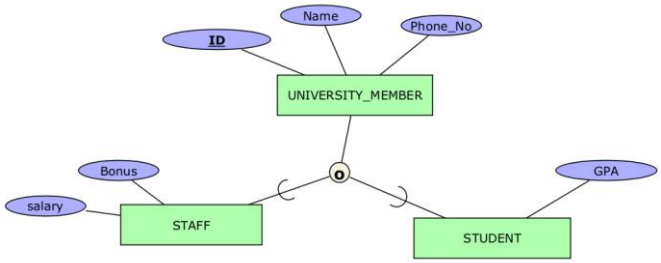
a)The University system has staff and students. Staff will receive Salary and bonus. Student's performance will be recorded with GPA. ID number, name and phone number will be recorded for every member of the University. A member can be staff as well as student at a time.



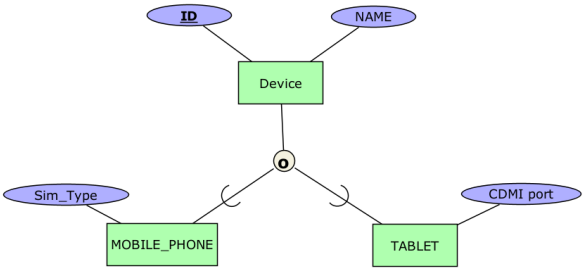
b) Redraw the EERD to reflect the following rule.

Sometimes, there can be members other than staff and students.

{OR } There are different type of members in university, of which some of them are STAFF and STUDENTS.



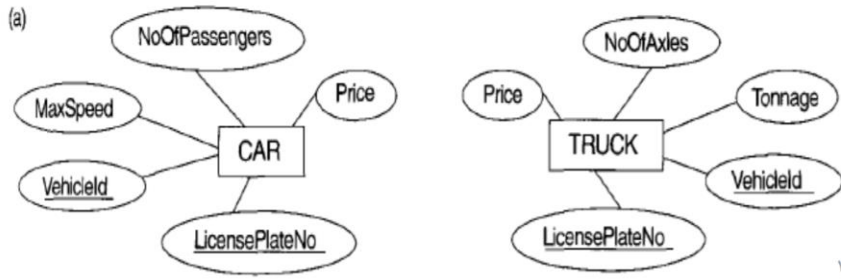
10. Write the business rule for the given EERD

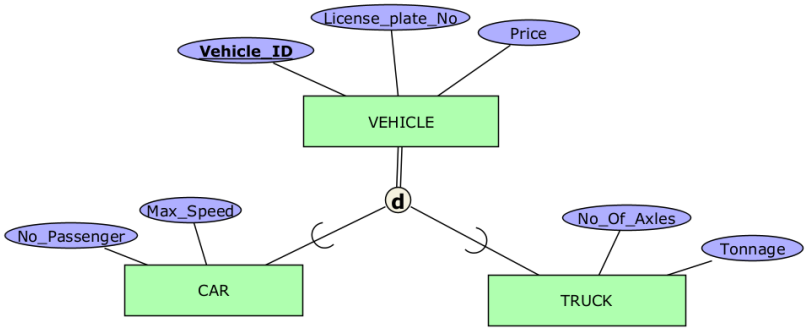
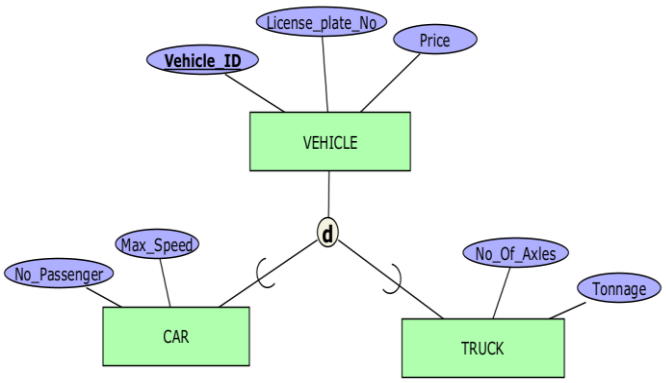


An electronic shop stores detail about different types of devices. Detail of Sim type will be stored for each mobile phone. Detail of CDMI port will be recorded for all tablets. All devices will have ID number and name. Some devices will work as mobile phone as well as tablets at the same time. (OR Mobile phone can work as Tablet and vice versa)

Sometimes, the device does not belong to mobile phone or tablet.

11. Apply generalization in the below model. Assume that there will be no vehicles other than car or truck in the model.





12. Consider the EERD that you developed to answer Q6 of this activity. Assume that there are some vehicles that are neither CAR nor TRUCK. (not belongs to both specialization). Which part of the diagram has to be modified?